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A Appendix

A.1 VQ-VAE dictionary updates with Exponential Moving Averages

As mentioned in Section 3.2, one can also use exponential moving averages (EMA) to update the dictionary items instead of the loss term from Equation 3:

$$\|\text{sg}[z_e(x)] - e\|_2^2. \tag{4}$$

Let $\{z_{i,1}, z_{i,2}, \dots, z_{i,n_i}\}$ be the set of n_i outputs from the encoder that are closest to dictionary item e_i, so that we can write the loss as:

$$\sum_j^{n_i} \|z_{i,j} - e_i\|_2^2. \tag{5}$$

The optimal value for e_i has a closed form solution, which is simply the average of elements in the set:

$$e_i = \frac{1}{n_i} \sum_j^{n_i} z_{i,j}.$$

This update is typically used in algorithms such as K-Means.

However, we cannot use this update directly when working with minibatches. Instead we can use exponential moving averages as an online version of this update:

$$N_i^{(t)} := N_i^{(t-1)} * \gamma + n_i^{(t)}(1 - \gamma) \tag{6}$$

$$m_i^{(t)} := m_i^{(t-1)} * \gamma + \sum_j z_{i,j}^{(t)}(1 - \gamma) \tag{7}$$

$$e_i^{(t)} := \frac{m_i^{(t)}}{N_i^{(t)}}, \tag{8}$$

with γ a value between 0 and 1. We found $\gamma = 0.99$ to work well in practice.